

Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

Elvis A. F. Martis,[†] Morgane Rousselle,[‡] Vincent Sauzeau,[‡] and Stéphane Téletchéa^{*,†}

[†]*US2B, Nantes Université, CNRS, UMR 6286, F-44000 Nantes, FRANCE*

[‡]*UMR_S 1087 / UMR_C 6291 l'Unité de Recherche de l'Institut du Thorax, F-44000 Nantes, FRANCE*

E-mail: stephane.teletchea@univ-nantes.fr

Supplementary Information

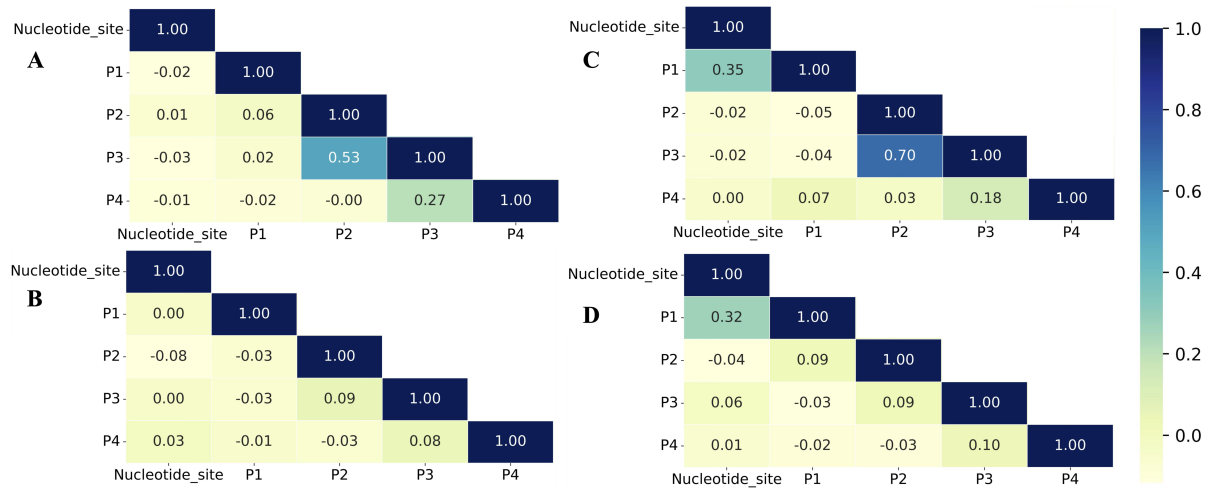


Figure S1: Pearson's Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding. (A) GDP-bound Rac1; (B) GTP-bound Rac1; (C) Nucleotide-free state created by deleting GDP from Rac1; (D) Nucleotide-free state created by deleting GTP from Rac1.

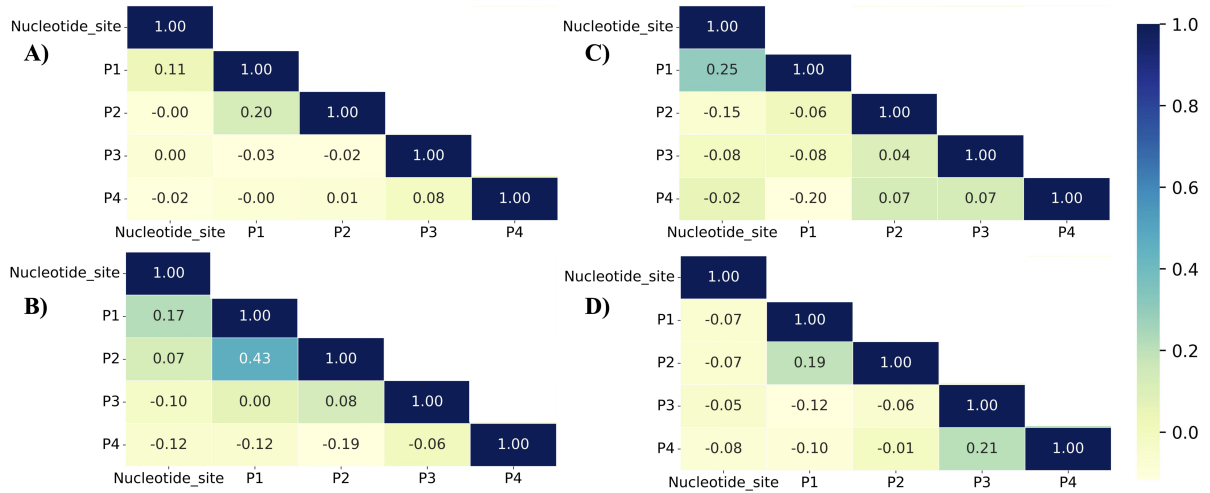


Figure S2: Pearson's Correlation for the volume fluctuations among different pockets and the effect of GEF binding. (A) PREX1-bound Rac1; (B) TIAM1-bound Rac1; (C) VAV1-bound Rac1; (D) DOCK2-bound Rac1. Rac1 in all GEF-bound systems is in the nucleotide-free state.

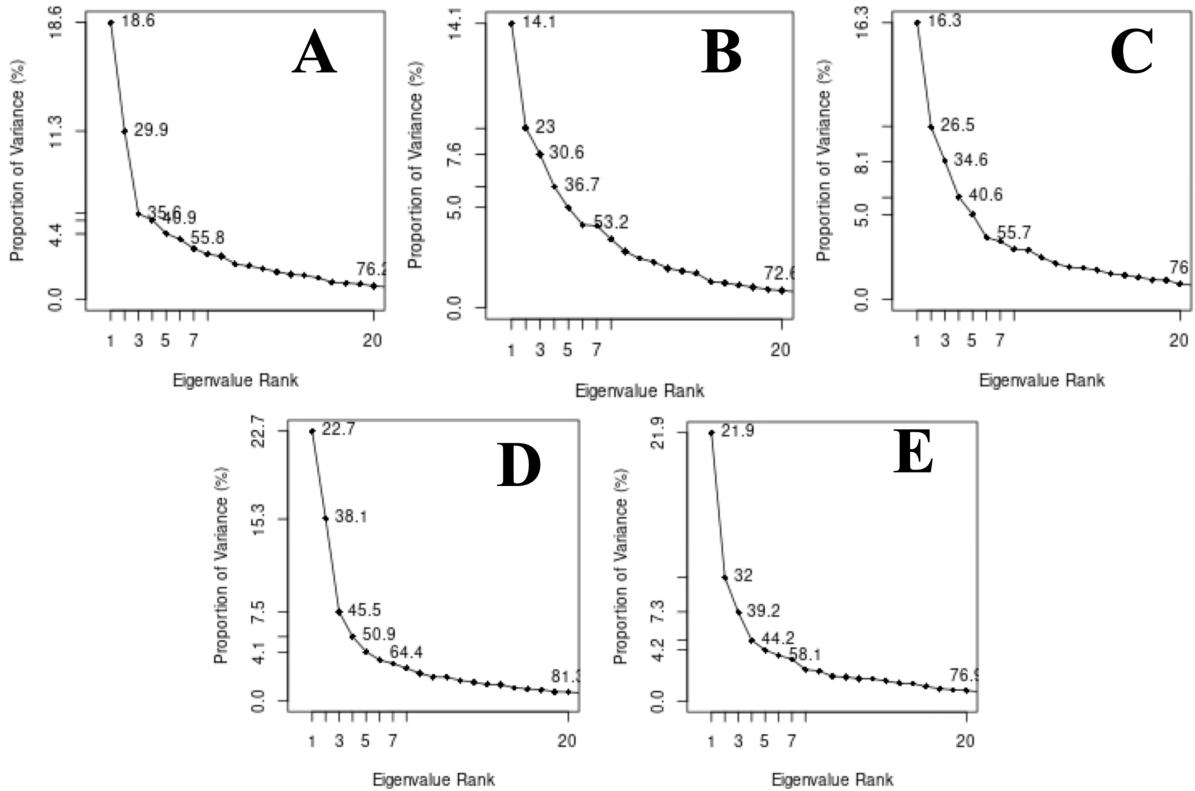


Figure S3: Scree plot to compute the cumulative variance explained by principal components. (A) *apo*-Rac1 in TIP3P water box; (B) GTP-bound Rac1 in TIP3P water box; (C) *apo*-Rac1 in TIP3P water+isopropanol box; (D) *apo*-Rac1 in TIP3P water+phenol box; (E) *apo*-Rac1 in TIP3P water+toluene box.

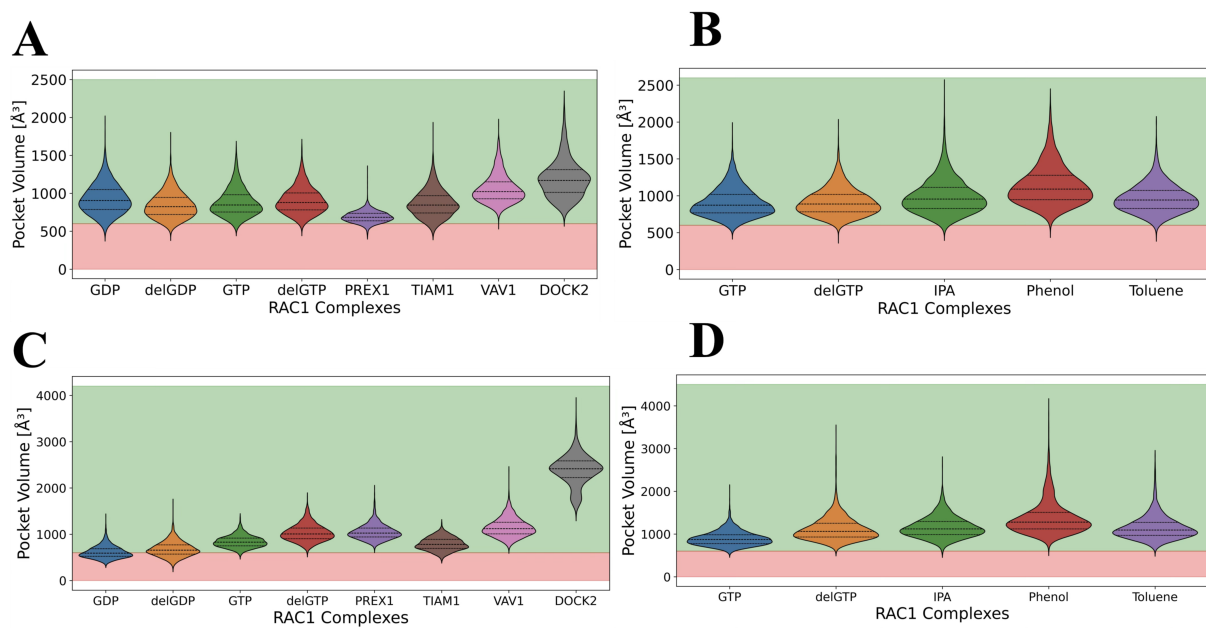


Figure S4: Analysis of pocket volume changes by combining pockets P1 with P2, and P3 with P4. **(A)** Combined pocket volume analysis for site P1+P2 ; **(B)** Combined pocket volume analysis for site P1+P2 in co-solvent; **(C)** Combined pocket volume analysis for site P3+P4; **(D)** Combined pocket volume analysis for site P3+P4 in co-solvent.

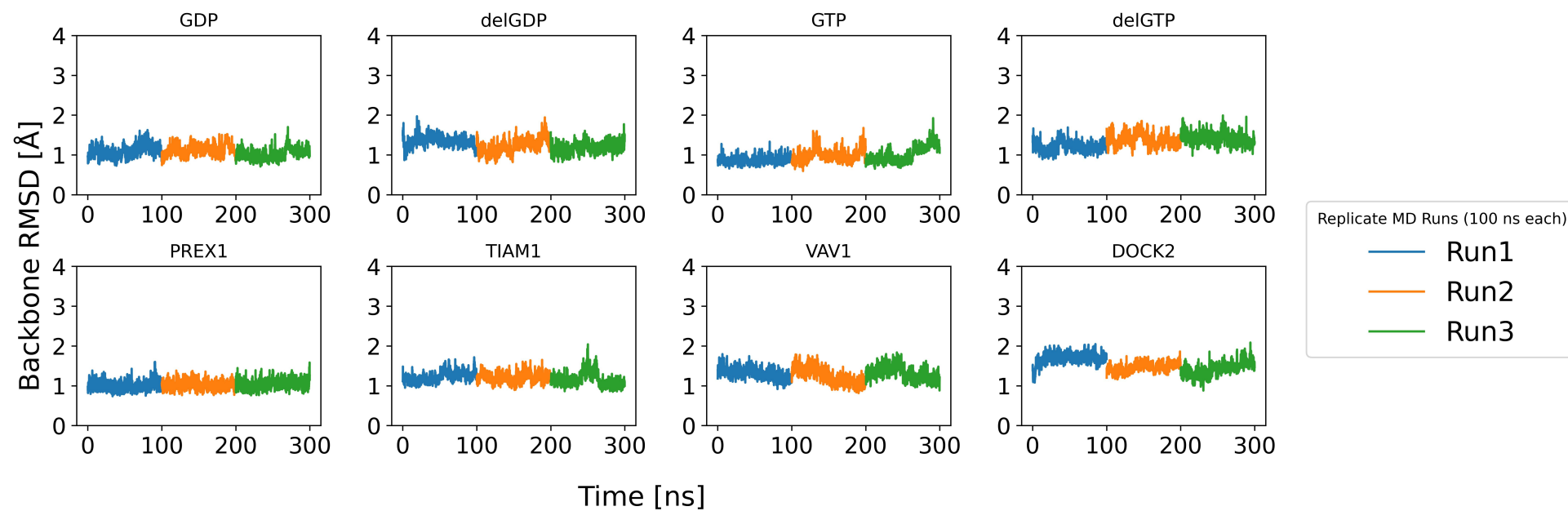


Figure S5: Time-evolution of RMSD of backbone atoms from Full Rac1 protein.

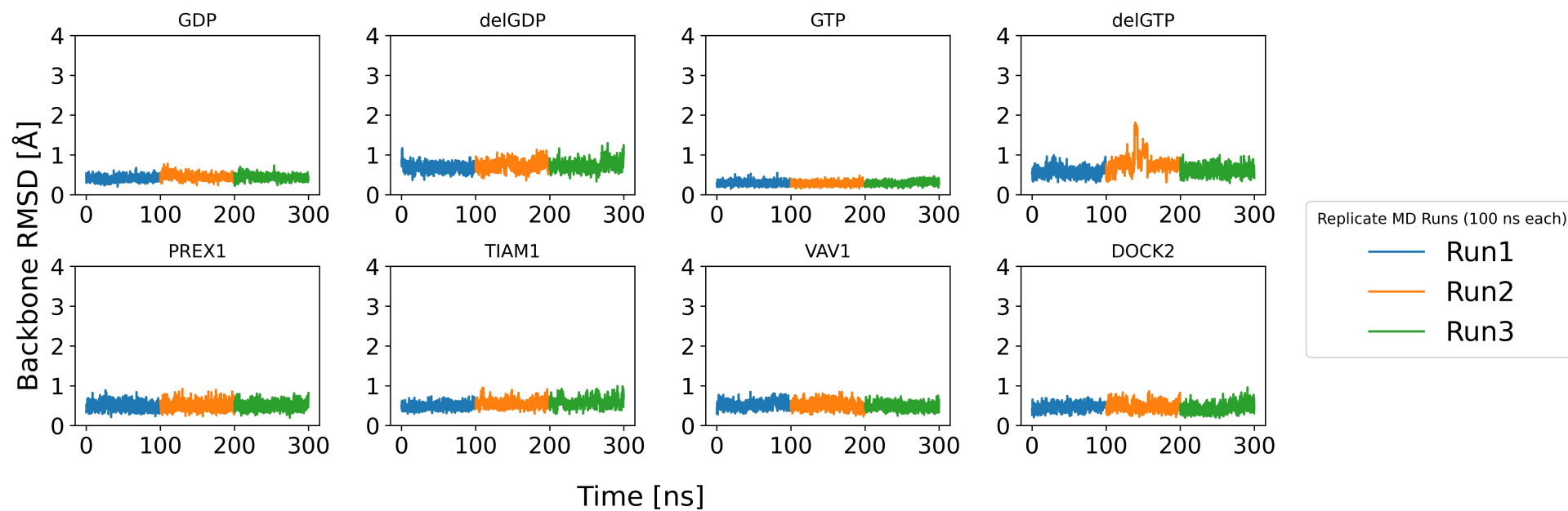


Figure S6: Time-evolution of RMSD of backbone atoms from the P-loop region.

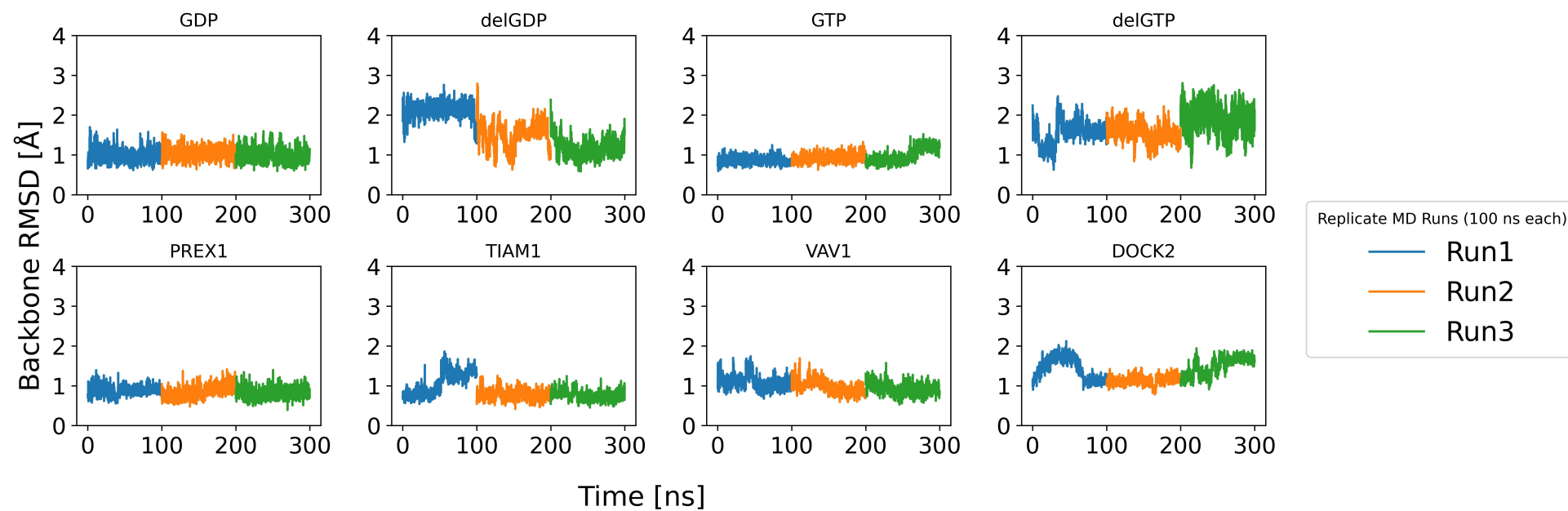


Figure S7: Time-evolution of RMSD of backbone atoms from the switch I region.

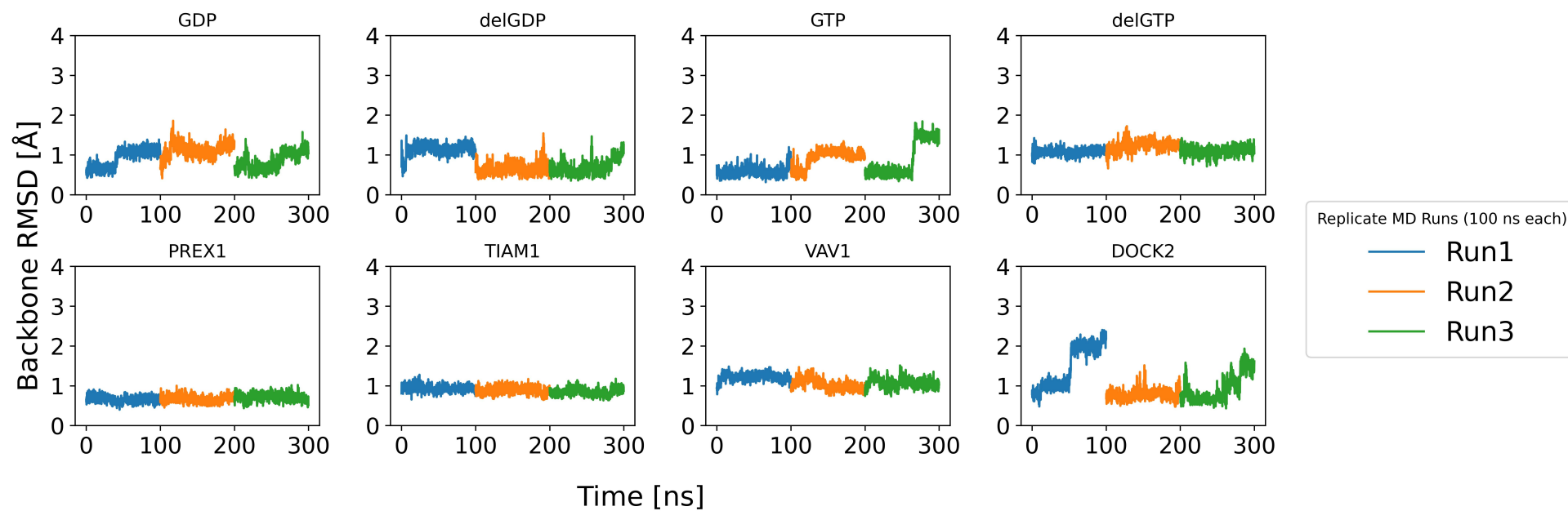


Figure S8: Time-evolution of RMSD of backbone atoms from the switch II region.